

RentSafe

Decentralized Rental Deposit Escrow Platform

From trust-based deposits to programmable agreements.

- Secure rental deposits with blockchain-powered Soroban smart contracts
- Transparent, automated settlement between tenants and landlords
- Structured refund and partial deduction approval workflows
- Built natively on Stellar for low-cost, instant transactions

Rental Deposits Are Built on Trust — Creating Friction & Risks for Both Parties

PAIN POINTS IN TODAY'S RENTAL MARKETS

- Tenants pay large security deposits directly to landlords without escrow protection.
- Deposit refunds are frequently delayed for weeks or months after lease completion.
- Disputes arise over property damage, cleaning costs, and repair deductions.
- Unclear lease conditions lead to subjective, unverified claims.
- Limited visibility into how deposits are held or invested during tenancy.
- Manual dispute resolution is slow, expensive, and stressful.
- Centralized platforms introduce additional intermediaries and transaction friction.
- Landlords face challenges recovering legitimate repair expenses fairly.

CORE PROBLEM

There is no neutral, transparent system that automatically enforces agreed deposit rules.

Trust-based deposits convert routine lease completions into friction, financial risk, and unnecessary legal overhead.

Programmable Escrow for Rental Deposits

HOW RENTSAFE SOLVES IT

- Deposit is locked into a Soroban smart contract upon agreement.
- Neither landlord nor tenant directly controls the escrowed funds.
- Rental agreement explicitly defines deposit, duration, and settlement rules.
- Funds remain locked safely throughout the entire rental lifecycle.
- Smart contract automatically enforces agreed settlement rules.
- Full refund executes automatically upon clean lease expiration.
- Partial deductions require tenant review and transparent sign-off.
- Every transaction creates a permanent, auditable blockchain record.

VALUE PROPOSITION

Tenant

Security, deposit safety & guaranteed refund rules without landlord holding funds.

Landlord

Reliable collection of verified repair deductions without payment default risk.

Property Manager

Centralized operational visibility & portfolio deposit tracking dashboard.

Platform

Trust-minimized rental infrastructure layer built for global scale.

Built Natively on Stellar & Soroban Smart Contracts

Fast Settlement

Rental deposits and refunds move in seconds. Perfect for time-sensitive housing transitions.

Low Transaction Costs

Sub-cent execution fees make frequent escrow calls and micro-deductions economically viable.

Soroban Smart Contracts

Programmable escrow logic, deterministic state machines, and permissioned contract actions.

Asset Flexibility

Support for XLM, USDC, and custom Stellar assets for stable real-world rental transactions.

Transparent Ledger

Immutable transaction history provides a tamper-proof audit log for all deposit agreements.

Future-Ready Ecosystem

Seamless integration with Stellar Anchors for fiat on/off-ramps and global financial rails.

How RentSafe Works (End-to-End Lifecycle)

01 Create Agreement

Landlord connects Stellar wallet and defines deposit amount, lease duration, and terms.

02 Lock Deposit

Tenant reviews terms and signs transaction. Deposit moves into Soroban escrow contract.

03 Rental Period

Escrow remains locked on-chain. Both parties monitor status with transparent history.

04 Lease Completion

Landlord submits a request for a full refund or an itemized partial deduction.

05 Tenant Review

Tenant reviews deduction request, accepts, or submits evidence for dispute review.

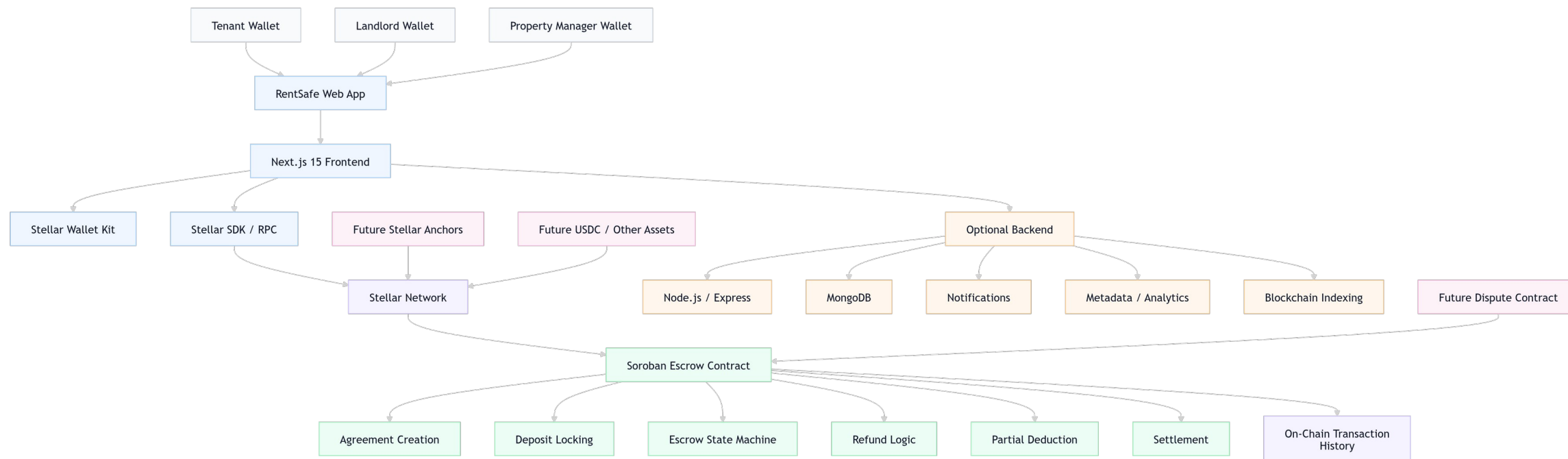
06 Settlement

Soroban contract automatically distributes funds: refund to tenant & approved deductions to landlord.

RentSafe Technical Architecture

06

On-chain escrow core with Next.js frontend, Stellar Wallet Kit & optional Node/Express backend.



KEY ARCHITECTURE PRINCIPLES

- Core escrow logic and deposit funds remain strictly on-chain in Soroban smart contracts.
- Stellar Wallet Kit provides secure user authentication and transaction signing.
- Frontend (Next.js 15) abstracts blockchain complexity into an intuitive Web app experience.
- Optional backend handles metadata, notifications, analytics, and blockchain indexing without holding custody.
- Modular design ready for future dispute-resolution contracts, Stellar Anchors, and USDC deposits.

Soroban Smart Contract Architecture & Security Focus

SOROBAN ESCROW CONTRACT LOGIC

- Creates rental agreements with stored landlord, tenant, deposit amount & timeline parameters.
- Locks deposited assets securely in contract custody during the lease period.
- Restricts state mutations strictly to authorized cryptographic participants.
- Validates escrow state transitions (Created → Locked → Pending → Settled).
- Processes full refunds automatically upon clean lease expiration.
- Executes partial deductions only upon explicit tenant review & approval.
- Prevents unauthorized fund releases, early withdrawals, or locked fund deadlocks.

SECURITY & SAFETY HIGHLIGHTS

- **Explicit Access Control:**
Strict role-based caller verification for every contract invocation.
- **State-Machine Validation:**
Ensures contract actions only occur in valid lifecycle states.
- **Safe Fund Custody:**
Soroban contract sandbox prevents unauthorized external calls or reentrancy.
- **Deterministic Settlement:**
Rule execution is entirely code-enforced without arbitrary intervention.
- **Minimal Off-Chain Dependency:**
Escrow funds and core state do not depend on external servers.
- **Security Audit Path:**
Testnet-first deployment with formal security review prior to mainnet.

Building Escrow Infrastructure for a Fragmented Rental Ecosystem

TARGET CUSTOMER SEGMENTS

- Individual landlords looking for deposit security
- Tenants wanting guaranteed refund protection
- Property management companies
- Real estate agencies & rental brokerages
- Digital rental marketplaces & platforms

MARKET PAIN POINTS

- Rental deposits are universal across global housing markets.
- Handling is fragmented and dependent on local manual processes.
- Disputes create heavy administrative & legal overhead.
- Centralized solutions charge high platform fee margins.

EXPANSION OPPORTUNITY

- Start with simple residential rental escrow.
- Expand into multi-property management dashboards.
- Commercial real estate rental agreements.
- API-based escrow infrastructure for property portals.
- Cross-border international rental deposit workflows.

RENTSAFE POSITIONING: The trusted decentralized escrow infrastructure layer for the global digital rental economy.

4-Phase Expansion Roadmap (From Developer MVP to Ecosystem Infrastructure)

PHASE 1

Developer & MVP Adoption

- Launch Stellar Testnet MVP
- Demonstrate complete escrow loop
- Partner with web3 communities
- Onboard early landlords & tenants
- Validate core UX & collect feedback

PHASE 2

Property Manager Adoption

- Target small property businesses
- Introduce multi-property dashboard
- Add automated notifications & audit logs
- Support portfolio deposit monitoring
- Drive repeat usage & property listings

PHASE 3

Enterprise Expansion

- Target real estate agencies & portals
- Offer white-label API integrations
- Introduce role-based permissions
- Add advanced analytics & reporting
- Prepare identity verification integrations

PHASE 4

Ecosystem Growth

- Support additional Stellar assets & USDC
- Integrate Stellar Anchors for fiat ramps
- Enable fiat-to-blockchain escrow UX
- Expand into international rental markets

Staged Progression: MVP → Platform → Mainnet Vision

LEVEL 4

MVP Launch

- Wallet authentication
- Rental agreement creation
- Deposit locking on Soroban
- Escrow state management
- Full refund workflow
- Tenant & Landlord dashboards
- Stellar Testnet deployment

LEVEL 5

Advanced Escrow

- Partial deduction requests
- Deduction approval workflow
- Activity timeline & audit log
- Property management dashboard
- Automated notification triggers
- Multiple active agreements support

LEVEL 6

Property Platform

- Multi-property management
- Multi-user role permissions
- Analytics and portfolio reporting
- Advanced escrow monitoring APIs
- Identity verification prep
- Contract security audit

LEVEL 7

Mainnet Vision

- Stellar Mainnet deployment
- Dispute resolution contracts
- Multi-signature approvals
- Stellar Anchor USDC integration
- Mobile native applications
- AI agreement summaries & OCR

Why RentSafe? Traditional Process vs RentSafe Escrow

TRADITIONAL RENTAL PROCESS	RENTSAFE ESCROW PLATFORM
Deposit held directly by landlord/intermediary	Deposit locked in secure Soroban smart contract
Trust-based, high dispute potential	Smart-contract enforced settlement rules
Manual, delayed refund process after move-out	Programmable automated refund execution
Limited visibility into funds during lease	Immutable on-chain transaction history
Disputes handled manually with high stress	Structured approval & deduction workflow
Centralized infrastructure with platform risk	Stellar-based decentralized escrow layer
High operational friction & admin overhead	Automated lifecycle reducing workload
Limited asset flexibility	Future USDC & multi-asset support

KEY DIFFERENTIATORS

Blockchain-native escrow • Transparent deposit lifecycle • Sub-cent Stellar cost • Designed for future dispute resolution contracts • API-ready for rental platform integration

Sustainable Monetization Streams & Platform Philosophy

POTENTIAL REVENUE STREAMS

- **Escrow Transaction Fee**
Small micro-fee collected per successfully settled deposit agreement.
- **Property Manager Subscriptions**
SaaS tier for property managers handling multi-unit portfolios.
- **Enterprise API Usage**
API access fees for large real estate platforms and rental portals.
- **White-Label Infrastructure**
Licensing RentSafe escrow rails to custom real estate platforms.
- **Marketplace Integration Fees**
Partnership revenue sharing with rental listing marketplaces.
- **Advanced Analytics & Reporting**
Premium reporting add-ons for institutional property managers.

MONETIZATION PRINCIPLE

Keep core deposit escrow affordable while monetizing high-value operational services.

By keeping deposit escrow low-cost, RentSafe maximizes adoption among tenants and landlords, while generating predictable SaaS & API revenue from property managers and enterprise partners.

The Future of Rental Deposits — From Trust to Programmable Trust

Programmable Escrow

Make rental deposits fully programmable, replacing manual holding and trust uncertainty.

Dispute Reduction

Drastically reduce unnecessary dispute cycles through structured, transparent approval rules.

Tenant Confidence

Improve tenant trust while ensuring landlords can reliably collect legitimate repair claims.

Platform Infrastructure

Connect digital rental marketplaces directly to decentralized escrow rails worldwide.

"RentSafe aims to become the trusted decentralized infrastructure layer for rental security deposits worldwide."

RentSafe

Secure Deposits. Transparent Agreements. Automated Settlement.

- Built on Stellar — Fast settlement & sub-cent transaction costs
- Powered by Soroban — Trustless, programmable escrow smart contracts
- Designed for Landlords, Tenants & Property Managers
- Starting with decentralized deposit escrow → Expanding toward global rental infrastructure

From Trust → To Programmable Trust